US 20250284229A1

(19) United States
(12) Patent Application Publication
YOSHIDA et al.

(10) Pub. No.: US 2025/0284229 A1
(43) Pub. Date: Sep. 11, 2025

(54) FIXING DEVICE

(71) Applicant: CANON KABUSHIKI KAISHA, Tokyo (JP)

(72) Inventors: KEISUKE YOSHIDA, Chiba (JP); HIDEKAZU TATEZAWA, Saitama (JP); YOUICHI CHIKUGO, Chiba (JP); YUYA YAMADA, Chiba (JP)

(21) Appl. No.: 19/061,244

(22) Filed: Feb. 24, 2025

(30) Foreign Application Priority Data

Mar. 8, 2024 (JP) 2024-035941

Publication Classification

(51) Int. Cl.
G03G 15/20 (2006.01)

(52) U.S. Cl.
CPC *G03G 15/2017* (2013.01); *G03G 15/2064* (2013.01); *G03G 2215/2038* (2013.01)

(57) ABSTRACT

A fixing device includes a belt unit including a rotating endless belt, a stretching roller, a first support plate, and a second support plate, a rotatable pressing member disposed outside the belt and forming a nip portion capable of sandwiching and conveying the recording material together with the belt, a frame supporting the first support plate and the second support plate, an adjustment unit provided on the frame and being capable of adjusting a position of the first support plate in a direction intersecting with the rotation axis direction, and a cover member covering the belt unit, the frame, and the adjustment unit. The cover member includes an opening portion formed at a position corresponding to the adjustment unit and communicating with an outside.

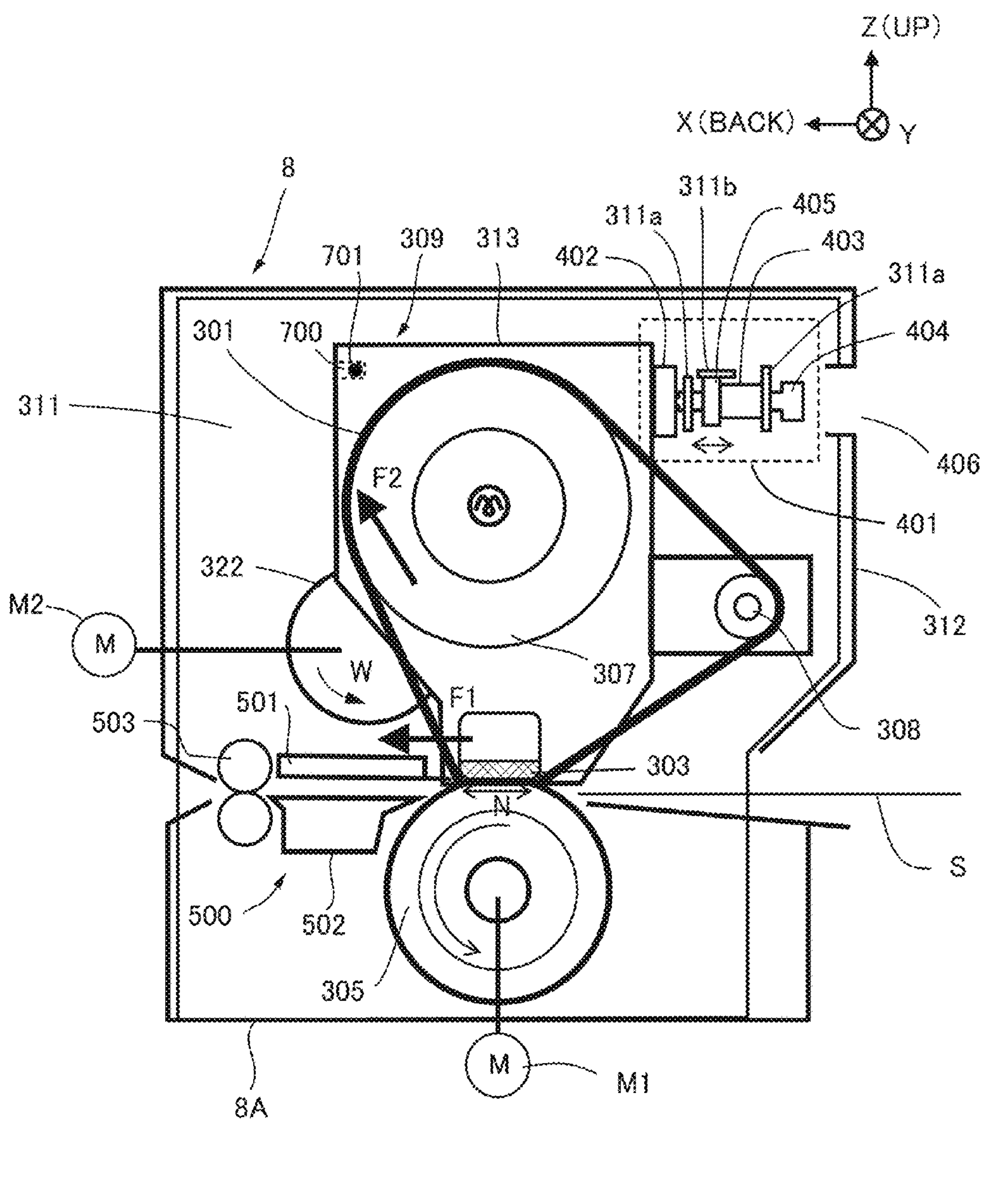


FIG.1

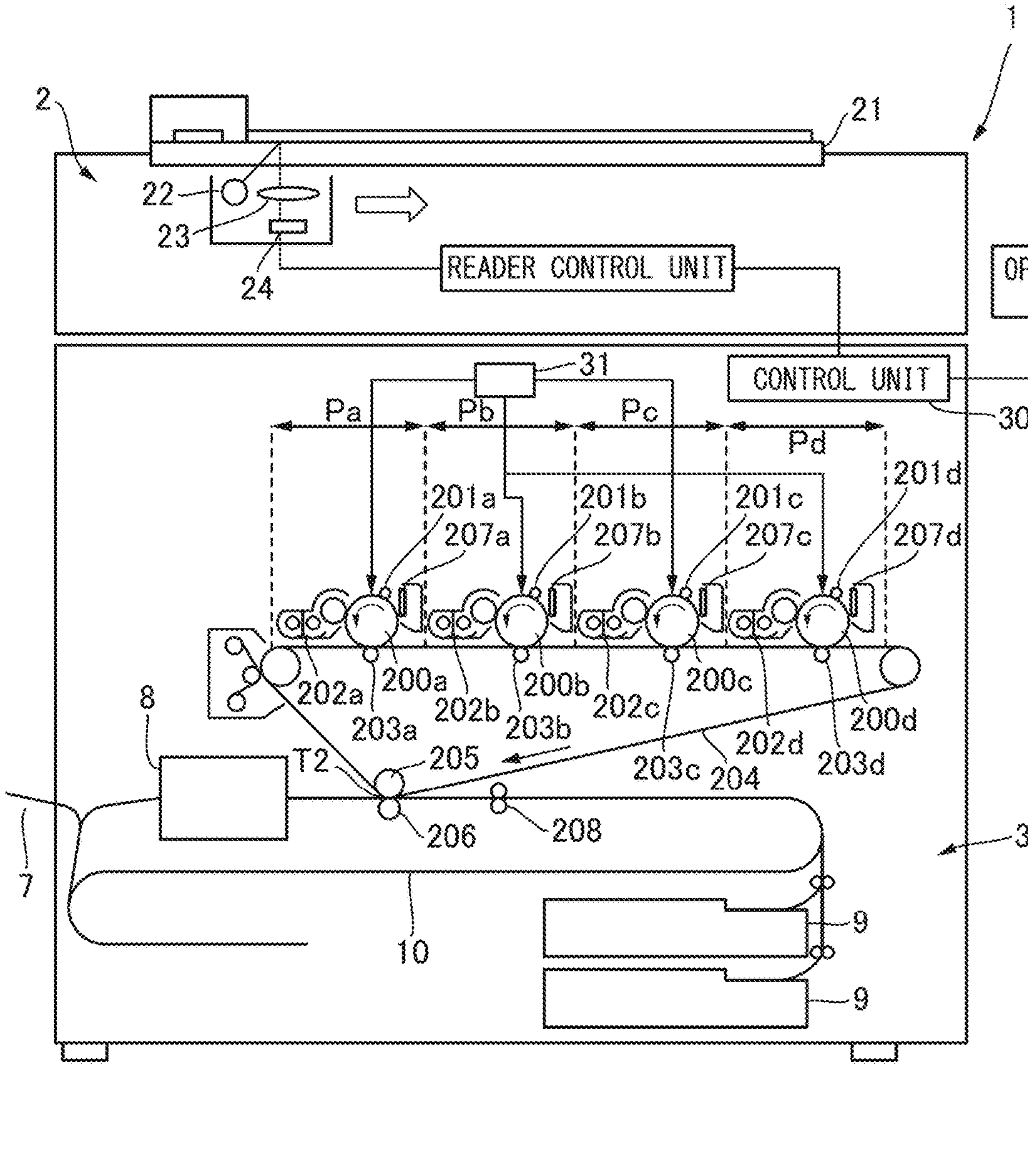

1
2
21
22
23
24
READER CONTROL UNIT
4
OPERATION UNIT
31
CONTROL UNIT
30
Pa
Pb
Pc
Pd
201a
207a
201b
207b
201c
207c
201d
207d
202a
200a
203a
202b
200b
203b
202c
200c
203c
202d
200d
203d
204
8
T2
205
206
208
7
10
3
9
9

FIG.2

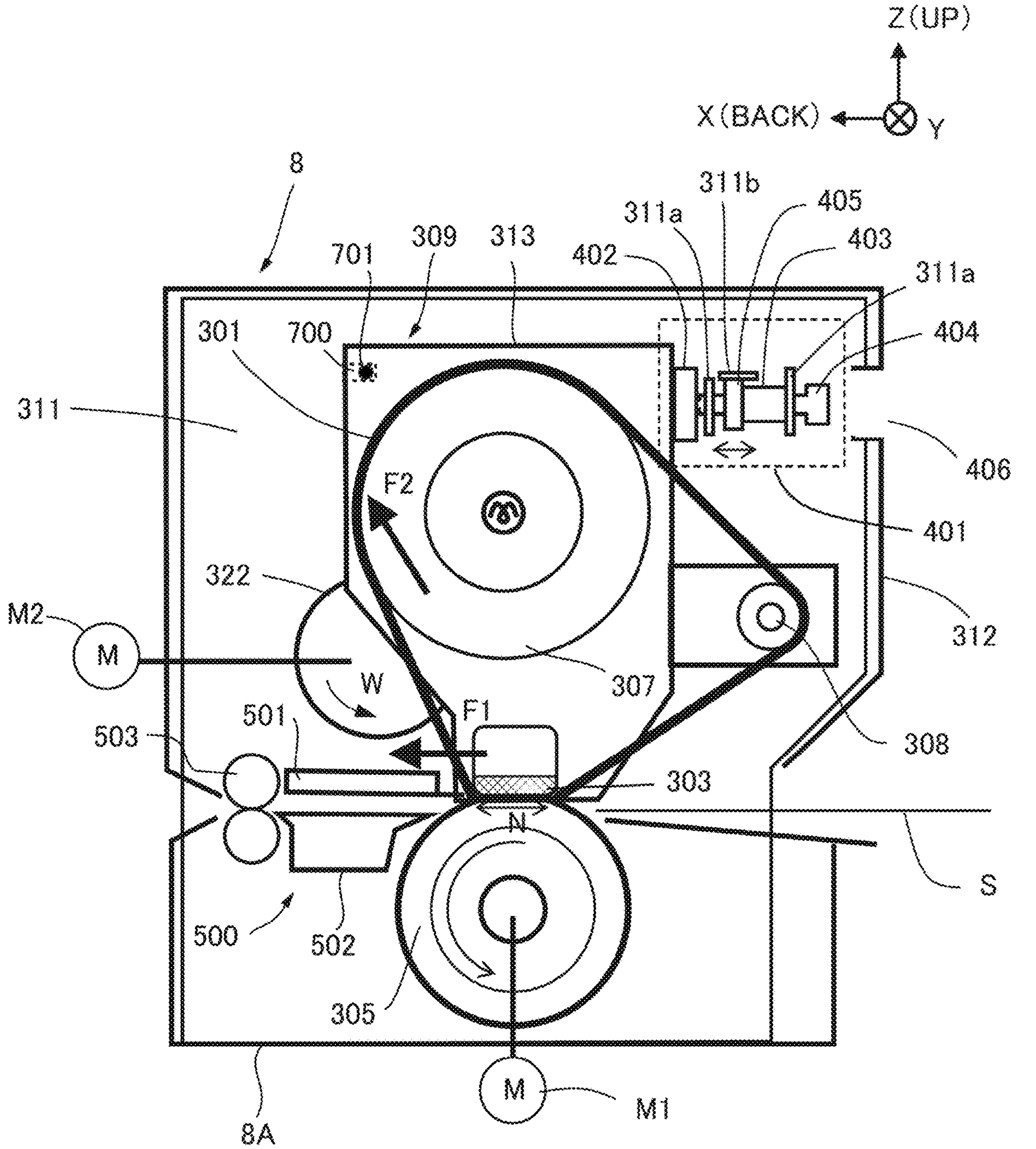

Z(UP)
X(BACK)
Y
8
701
309
313
311b
311a
405
402
403
311a
301
700
404
311
406
F2
401
322
312
M2
M
W
307
F1
501
308
503
303
N
S
500
502
305
M
M1
8A

FIG.3

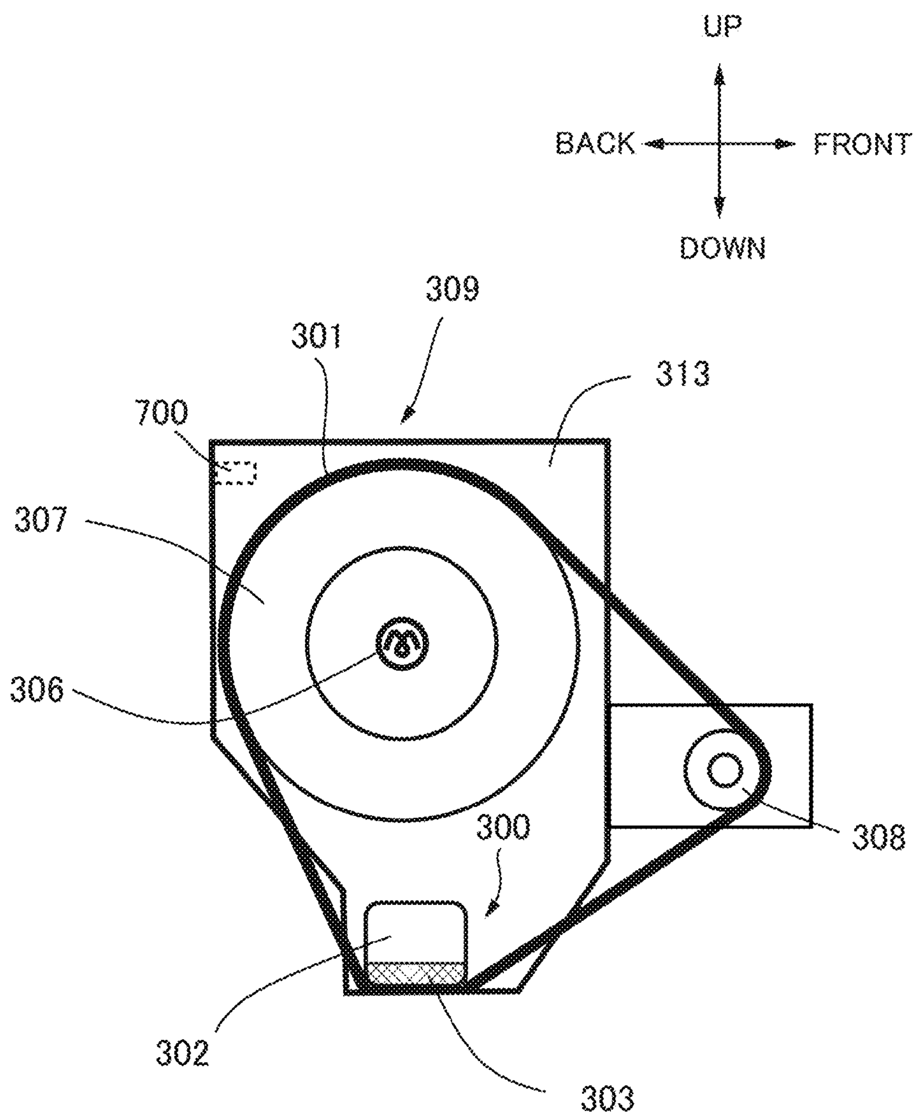


FIG.4

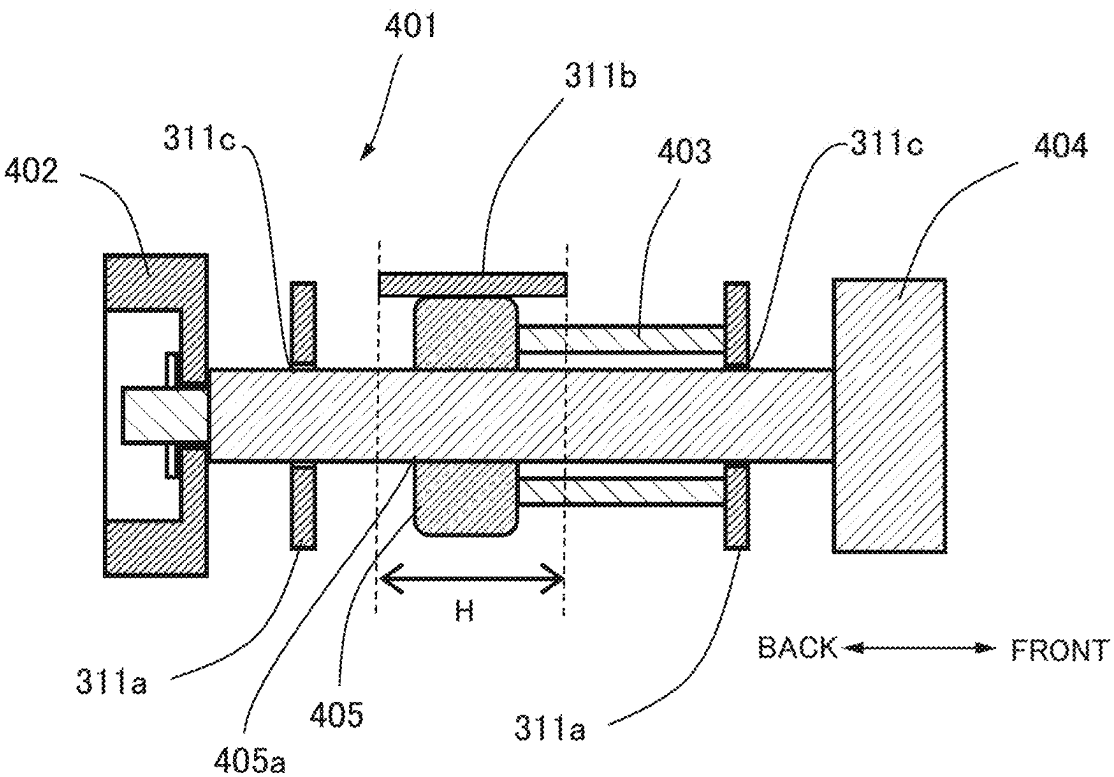
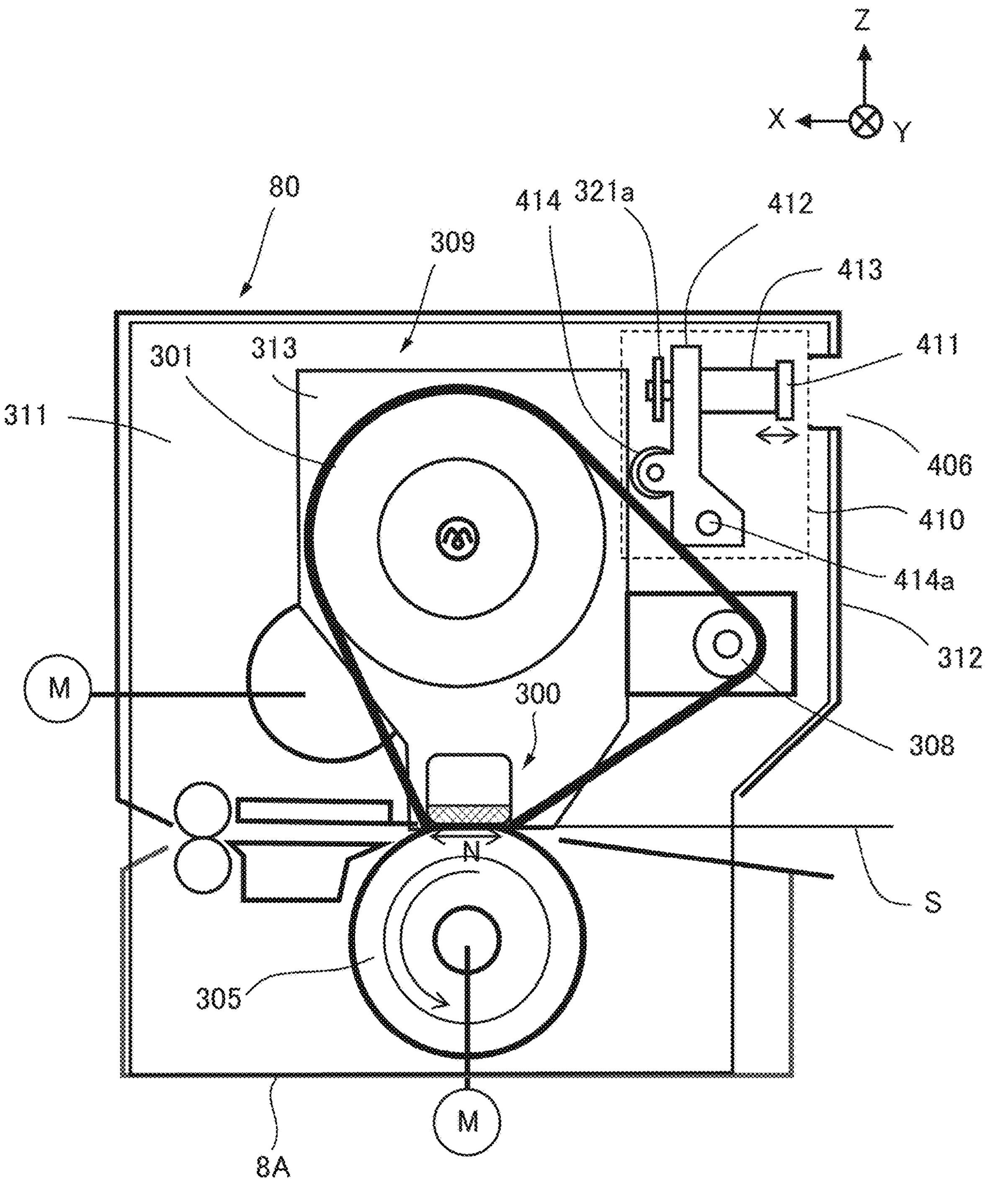


FIG.5



Z
X
Y
80
309
414
321a
412
413
411
406
410
414a
312
308
311
301
313
M
300
N
S
305
M
8A

FIG.6

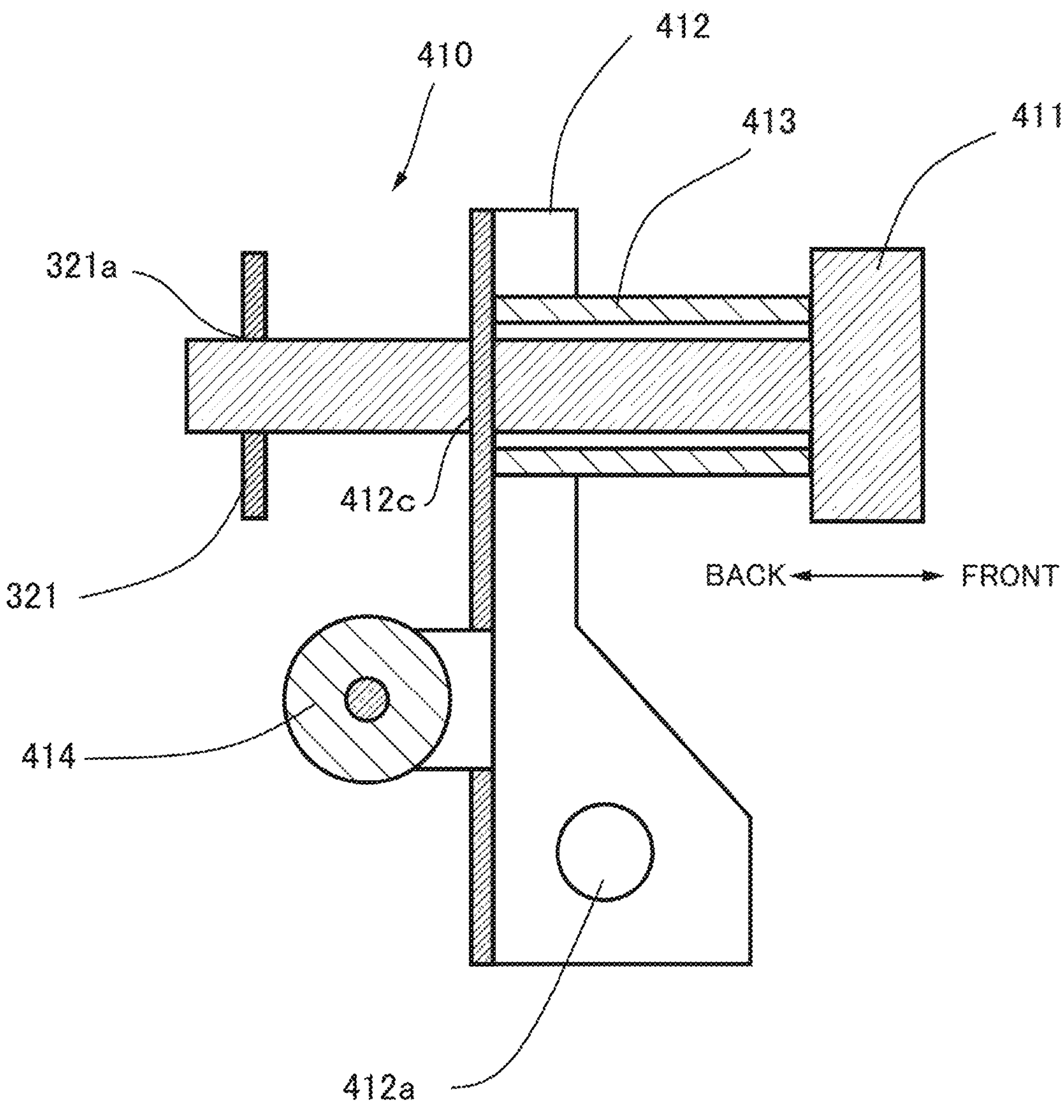


FIG.7

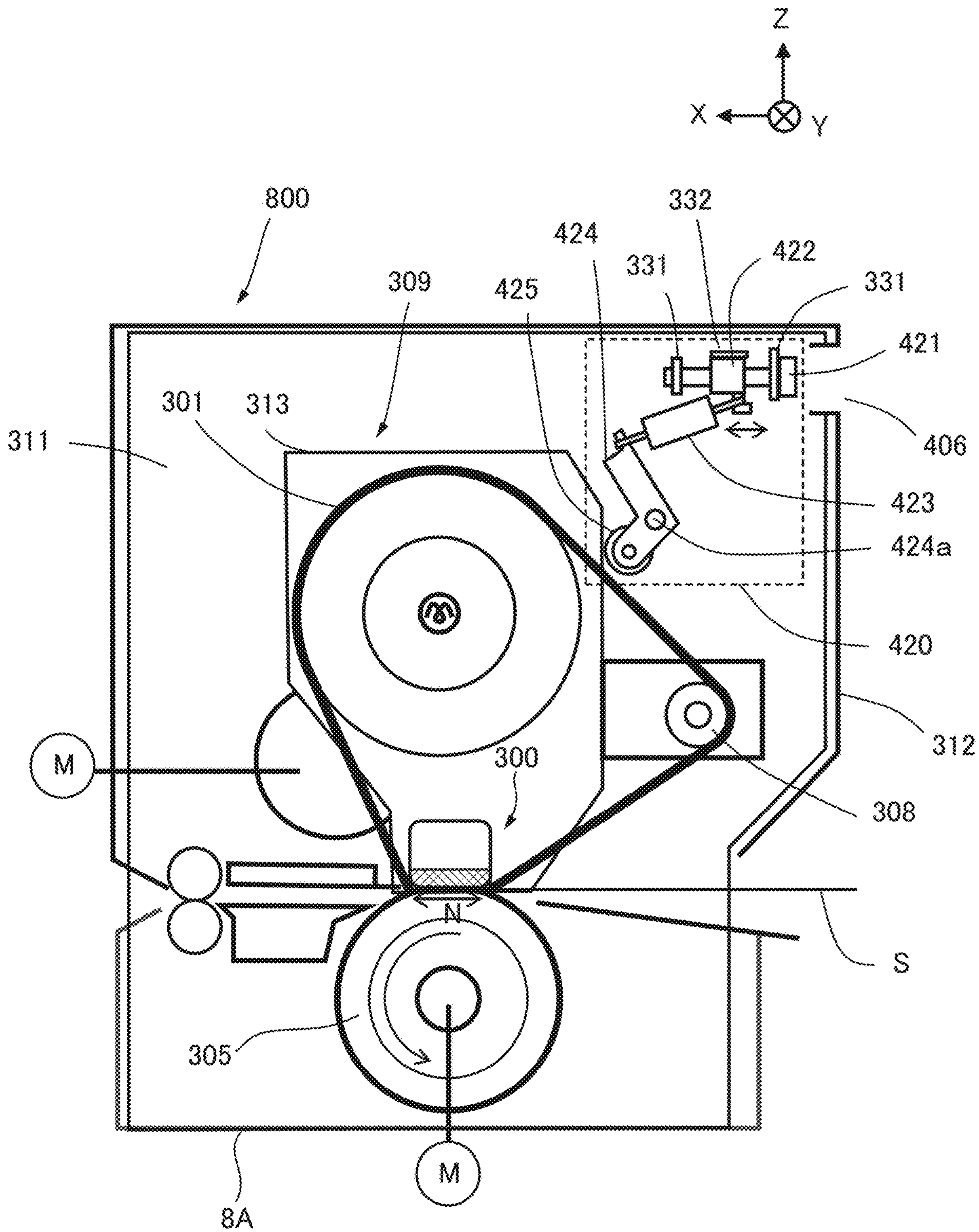

Z
X
Y
800
332
424
422
331
331
309
425
421
301
313
406
311
423
424a
420
312
M
300
308
N
S
305
M
8A

FIG.8

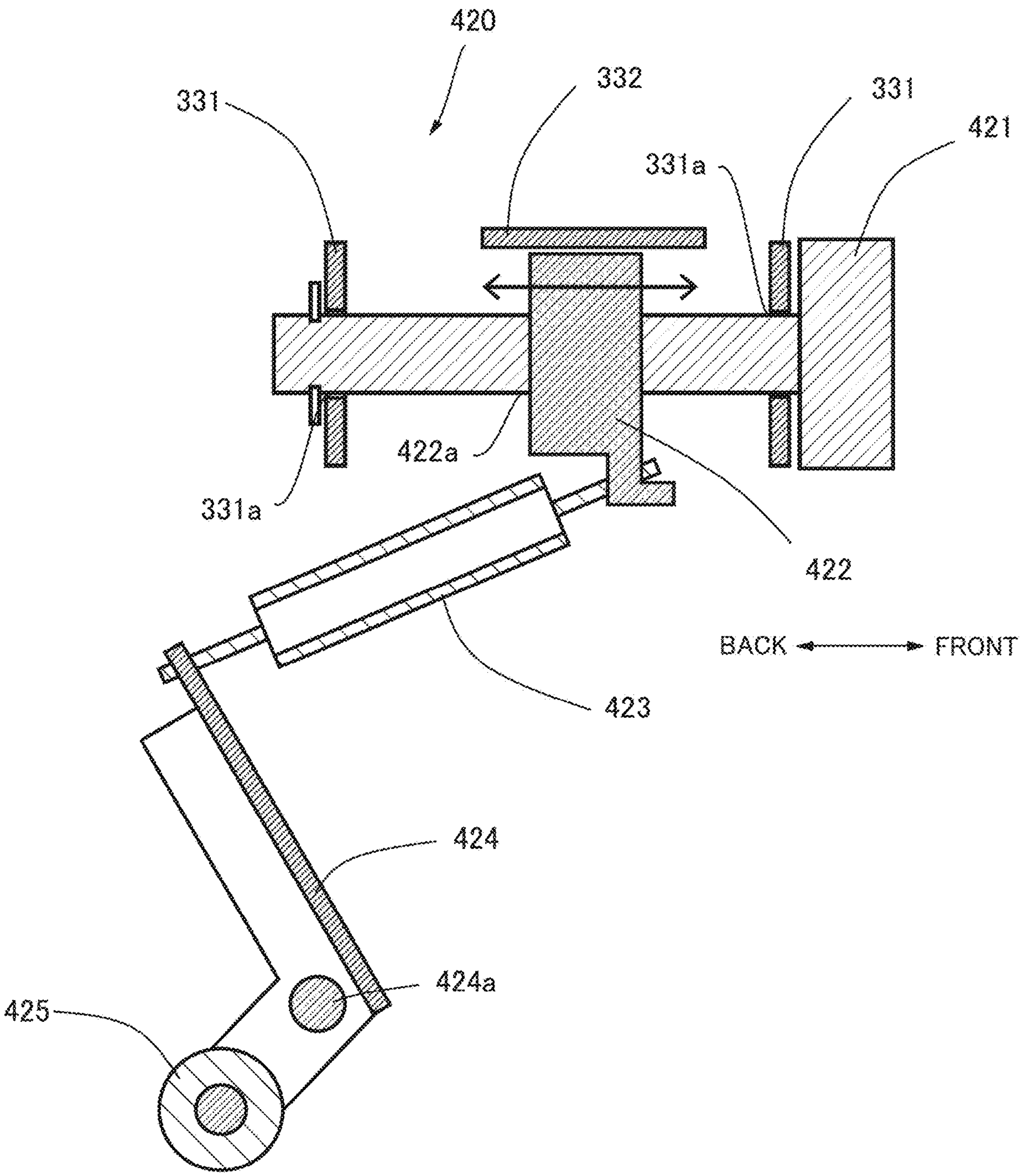

420
331
332
331
421
331a
422a
331a
422
BACK
FRONT
423
424
424a
425

FIG.9

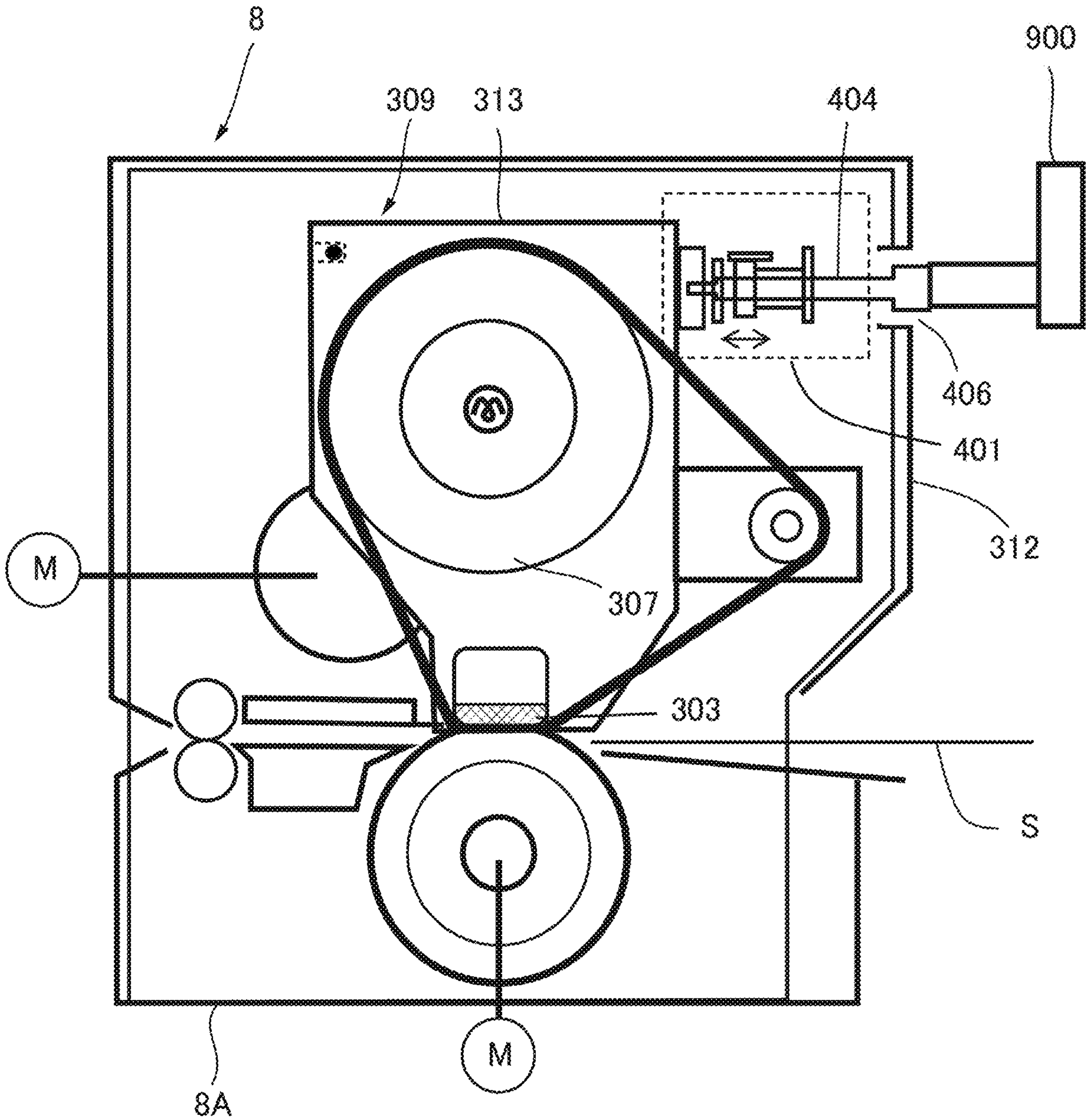
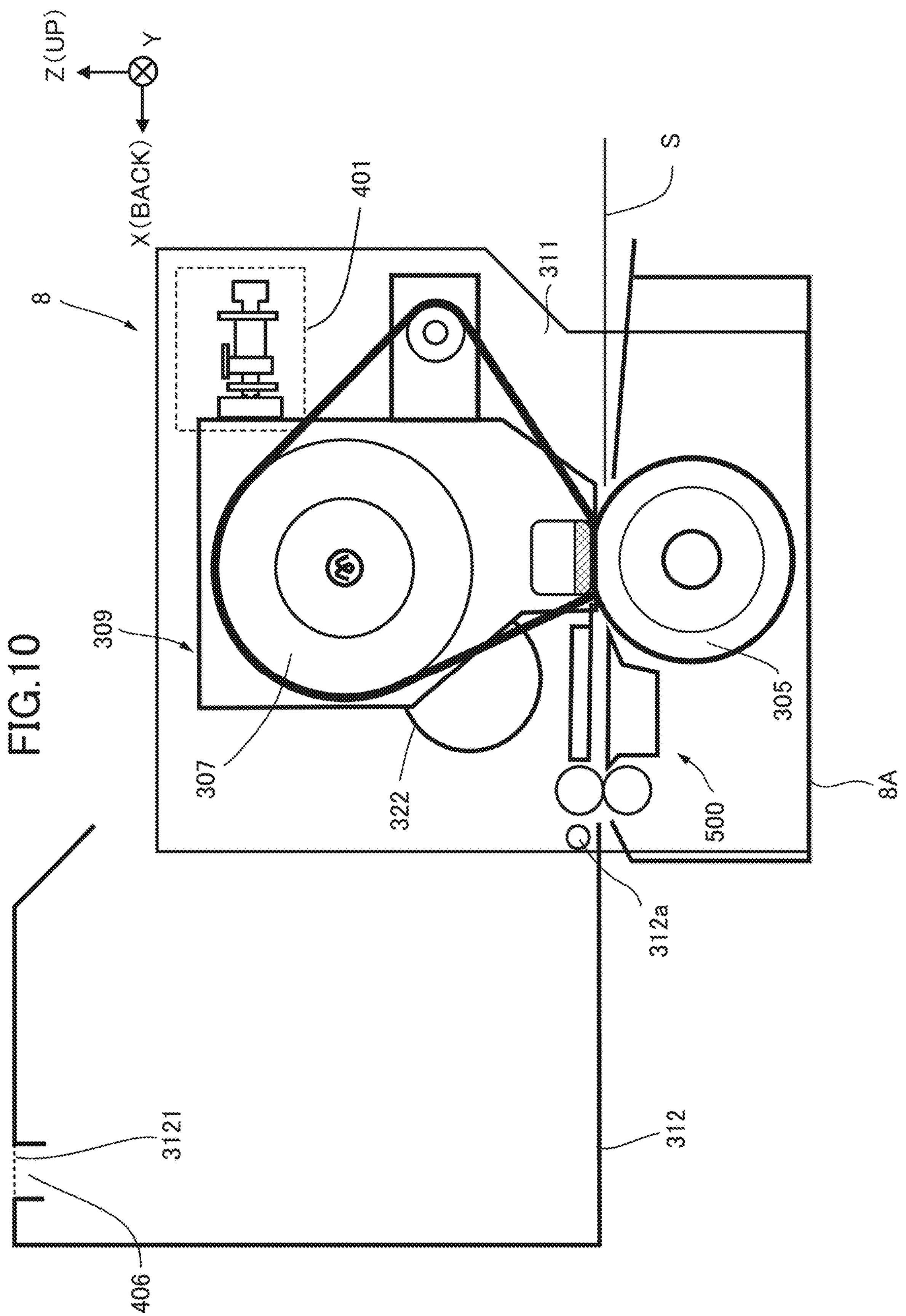


FIG.10



FIXING DEVICE

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a fixing device suitable for use in an image forming device such as a printer, a copier, a facsimile, or a multifunction machine.

Description of the Related Art

[0002] In an image forming device, a toner image that is not fixed to a recording material is formed, and the toner image is heated and pressurized by a fixing device to be fixed to the recording material. As an example of the fixing device, Japanese Patent Application Laid-Open Publication No. 2008-33227 proposes a belt heating-type fixing device using a belt. In the case of the belt heating type, in a case where the rotation axis directions of stretching rollers that stretch a belt are not parallel and are out of sync (referred to as an alignment error), belt tracking in which the rotating belt moves in the width direction intersecting with the rotation direction aside the end portions of the stretching rollers may occur. In a case where belt tracking occurs, the moved belt may come into contact with other members, and the belt or other members may be damaged. Therefore, steering control of adjusting belt tracking by detecting the widthwise position of the belt end portion by a sensor and swinging one of the stretching rollers (referred to as a steering roller) based on the detection result to change the relative angle of the steering roller with respect to the other stretching roller is performed.

SUMMARY OF THE INVENTION

[0003] According to one aspect of the present invention, a fixing device that fixes a toner image formed on a recording material to the recording material, includes a belt unit including a rotating endless belt, a stretching roller disposed inside the belt and configured to stretch the belt, a first support plate supporting one end portion of the stretching roller in a rotation axis direction, and a second support plate supporting the other end portion of the stretching roller in the rotation axis direction, a rotatable pressing member disposed outside the belt and forming a nip portion capable of sandwiching and conveying the recording material together with the belt, a frame supporting the first support plate and the second support plate, an adjustment unit provided on the frame and being capable of adjusting a position of the first support plate in a direction intersecting with the rotation axis direction, and a cover member covering the belt unit, the frame, and the adjustment unit. The cover member includes an opening portion formed at a position corresponding to the adjustment unit and communicating with an outside.

[0004] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a schematic view showing an image forming device suitable for using a fixing device of the present embodiment.

[0006] FIG. **2** is a schematic view showing a fixing device according to a first embodiment.

[0007] FIG. **3** is a schematic view showing a belt unit.

[0008] FIG. **4** is a cross-sectional view showing a pressing mechanism of the first embodiment.

[0009] FIG. **5** is a schematic view showing a fixing device of a second embodiment.

[0010] FIG. **6** is a cross-sectional view showing a pressing mechanism of the second embodiment.

[0011] FIG. **7** is a schematic view showing a fixing device of a third embodiment.

[0012] FIG. **8** is a cross-sectional view showing a pressing mechanism of the third embodiment.

[0013] FIG. **9** is a schematic view showing a case where a grip portion is provided in the pressing mechanism.

[0014] FIG. **10** is a schematic view showing a case where a cover member is in an open position.

DESCRIPTION OF THE EMBODIMENTS

First Embodiment

Image Forming Device

[0015] The present embodiment will be described. First, the configuration of an image forming device suitable for using a fixing device of the present embodiment will be described using FIG. **1**. An image forming device **1** is an electrophotographic full-color printer including four image forming units Pa, Pb, Pc, and Pd provided to correspond to four colors of yellow, magenta, cyan, and black. The present embodiment is a tandem type image forming device **1** in which the image forming units Pa, Pb, Pc, and Pd are disposed along a rotation direction of an intermediate transfer belt **204**. The image forming device **1** forms a toner image (image) on a recording material according to an image signal from an external host device such as a print server connected in a communicable manner to a document reading device **2** connected to a device main body **3** of the image forming device **1** or the device main body **3**. Examples of the recording material include sheet materials such as paper, a plastic film, and cloth.

[0016] As shown in FIG. **1**, the image forming device **1** includes the document reading device **2** and the device main body **3**. The document reading device **2** is a device configured to read a document placed on a document table glass **21**, light radiated from a light source **22** is reflected by the document, and an image is formed on a CCD sensor **24** via an optical system member **23** such as a lens. In a case where such an optical system unit has been scanned in a direction of an arrow under the control by a reader control unit, the optical system unit reads the document line by line and converts the document into electric signal data strings. An image signal obtained by the CCD sensor **24** is sent to the device main body **3**, and image processing corresponding to each image forming unit described later is performed by a control unit **30**. In addition, the control unit **30** also receives an external input from an external host device such as a print server as an image signal.

[0017] The device main body **3** includes a plurality of image forming units Pa, Pb, Pc, and Pd, and image formation is performed in each image forming unit based on the image signal described above. That is, the image signal is converted into a PWM (pulse width modulation-controlled) laser beam by the control unit **30**. A polygon scanner **31** serving as an exposure device scans the laser beam according to the image signal. Then, photosensitive drums **200*a*** to

200*d* serving as image carriers of the image forming units Pa to Pd are irradiated with laser beams.

[0018] The image forming unit Pa forms a yellow color (Y) toner image, the image forming unit Pb forms a magenta color (M) toner image, the image forming unit Pc forms a cyan color (C) toner image, and the image forming unit Pd forms a black color (Bk) toner image. Since these image forming units Pa to Pd have substantially the same configuration, the image forming unit Pa configured to form a yellow color (Y) toner image will be described below as an example, and the description of the other image forming units Pb to Pd will be omitted. In the image forming unit Pa, a toner image is formed on the surface of the photosensitive drum **200***a* based on the image signal.

[0019] A charging roller **201***a* charges the surface of the photosensitive drum **200***a* to a predetermined potential to prepare for the formation of an electrostatic latent image. An electrostatic latent image is formed on the surface of the photosensitive drum **200***a* charged to a predetermined potential by the laser beam from the polygon scanner **31**. A developer **202***a* develops the electrostatic latent image on the photosensitive drum **200***a* to form a toner image. A primary transfer roller **203***a* performs discharge from the back surface of an intermediate transfer belt **204**, applies a primary transfer bias having a polarity opposite to that of a toner, and transfers the toner image on the photosensitive drum **200***a* onto the intermediate transfer belt **204**. The surface of the photosensitive drum **200***a* after the transfer is cleaned by a cleaner **207***a*.

[0020] In addition, the toner image on the intermediate transfer belt **204** moves to the next image forming unit, toner images of respective colors formed by the image forming units Pa to Pd sequentially in the order of Y, M, C, and Bk are transferred, and an image of four colors is formed on the surface of the intermediate transfer belt **204**. Then, the toner image that has passed through the image forming unit Pd present most downstream in the rotation direction of the intermediate transfer belt **204** moves to a secondary transfer portion T**2** composed of a pair of secondary transfer rollers **205** and **206**. Then, in the secondary transfer portion T**2**, the toner image is secondarily transferred from the intermediate transfer belt **204** to a recording material by applying a secondary transfer electric field having a polarity opposite to the polarity of the toner image on the intermediate transfer belt **204**.

[0021] The recording material is accommodated in a cassette **9**, and the recording material fed from the cassette **9** is conveyed to, for example, a registration unit **208** composed of a pair of registration rollers and waits at the registration unit **208**. Thereafter, a timing is controlled by the registration unit **208** to align the toner image on the intermediate transfer belt **204** with the position of paper, and the recording material is conveyed to the secondary transfer portion T**2**.

[0022] The recording material onto which the toner image has been transferred in the secondary transfer portion T2 is conveyed to a fixing device **8**, and the toner image is fixed to the recording material by applying heat and pressure in the fixing device **8**. The recording material that has passed through the fixing device **8** is discharged to a discharge tray **7**. In a case where images are formed on both surfaces of the recording material, when the transfer and fixing of a toner image to a first surface (front surface) of the recording material are completed, the front and back of the recording material are reversed through a reverse conveying unit **10**, the transfer and fixing of a toner image to a second surface (back surface) of the recording material are performed, and the recording material is discharged onto the discharge tray **7**.

[0023] The control unit **30** controls the entire image forming device **1**. The control unit **30** has a central processing unit (CPU), a read only memory (ROM), and a random access memory (RAM). The CPU controls each unit while reading out a program corresponding to a control procedure stored in the ROM. The RAM stores work data or input data, and the CPU performs control with reference to data stored in the RAM based on the above-described program or the like. The control unit **30** can perform various settings and the like based on an input from an operation unit **4** in the image forming device **1**. The operation unit **4** may be, for example, a touch panel, and receives start inputs of various programs such as an "image forming job" and various data inputs according to a user's touch operation.

Fixing Device

[0024] Next, the fixing device **8** of the present embodiment will be described using FIG. **2** and FIG. **3**. In the present embodiment, a belt heating-type fixing device **8** using an endless fixing belt **301** is adopted. In FIG. **2**, "X" indicates a conveying direction of a recording material S, "Y" indicates a width direction, and "Z" indicates a pressurizing direction. In the present specification, the width direction (left-right direction) is a direction intersecting with the conveying direction of the recording material S in a fixing nip portion N described later, in other words, a rotation axis direction of a pressing roller **305**. In addition, a case where the fixing device **8** is viewed from the upstream side in the conveying direction is referred to as "front (or front surface)", a side opposite to the front surface is referred to as "back", the left side as viewed from the front surface is referred to as "left", and the right side as viewed from the front surface is referred to as "right". FIG. **2** and FIG. **3** show the fixing device **8** and a belt unit **309** as viewed from the left.

[0025] As shown in FIG. **2**, the fixing device **8** includes a fixing frame **8**A, the belt unit **309**, the pressing roller **305**, a discharge unit **500**, and a pressing mechanism **401**. Although not shown, the fixing frame **8**A is composed of pillars extending in the up-down direction at four corners, connection stays that connect the pillars each other in the left-right direction (width direction) or the front-back direction, and the like. Frame side plates **311** constituting a part of the fixing frame **8**A are disposed at both ends of the fixing frame **8**A in the left-right direction. The frame side plates **311** support the belt unit **309**, the pressing roller **305**, and the discharge unit **500**. In addition, a cover member **312** is installed on the fixing frame **8**A to cover the belt unit **309**. The cover member **312** is screwed to the fixing frame **8**A in a closed state.

[0026] Here, the cover member **312** is provided to be movable between a closed position in a closed state where the belt unit **309** is not exposed shown in FIG. **2** and an open position in an open state where the belt unit **309** is exposed as shown in FIG. **10**. The cover member **312** is screwed with a screw (not shown) to the fixing frame **8**A at the closed position in the closed state, and the fixing device **8** can perform fixing operation in a state where the cover member **312** remains screwed. Incidentally, in the case of maintenance work of the belt unit **309** or the like of the fixing device **8**, the belt unit **309** must be exposed. Therefore, the cover member **312** is provided to be movable from the closed position to the open position in a state where the screw has been removed. For example, as shown in FIG. **10**, the cover member **312** is provided such that an operator can open the cover member **312** from the front side toward the back side around a rotational movement axis **312***a* disposed downstream of the discharge unit **500** in the conveying direction of the recording material S as a rotational movement center.

Pressing Roller

[0027] The pressing roller **305** serving as a rotatable pressing member is rotatably pivoted by the left and right frame side plates **311** at both end portions in the width direction so as to be capable of pressurizing the fixing belt **301** of the belt unit **309** while abutting on the fixing belt **301**. A gear (not shown) is fixed to one end portion of the pressing roller **305**, and the pressing roller **305** is connected to a motor Ml via the gear and driven to rotate.

[0028] The pressing roller **305** is a roller in which a cylindrical core metal, an elastic layer on the outer circumference of the core metal, and furthermore, a release layer on the elastic layer are formed. As an example, a stainless steel (SUS) having a diameter of "72 mm" is used as the core metal, a conductive silicone rubber having a thickness of "8 mm" is used as the elastic layer, and the release layer is formed using a tetrafluoroethylene-perfluoroalkoxyethylene copolymer resin (PFA) having a thickness of "100 μm".

[0029] The pressing roller **305** abuts on the outer peripheral surface of the fixing belt **301** so as to sandwich the fixing belt **301** between a pressurizing pad **303** of the belt unit **309** and the pressing roller **305**, and forms the fixing nip portion N where the recording material S can be sandwiched and conveyed to fix the toner image on the recording material S. For this purpose, the pressing roller **305** is provided to be movable toward the pressurizing pad **303** by a driving source not shown. In the case of the present embodiment, the pressing roller **305** presses the fixing belt **301** such that, for example, the pressurizing force in the fixing nip portion N at the time of image formation becomes "1600 N", the length of the fixing nip portion N in the conveying direction becomes "24.5 mm", and the length of the fixing nip portion N in the width direction becomes "326 mm". The belt unit **309** receives a force corresponding to a driving force of the pressing roller **305** in a direction of an arrow F**1** in the fixing nip portion N.

Discharge Unit

[0030] The discharge unit **500** for discharging the recording material S that has passed through the fixing nip portion N to the outside of the fixing frame **8**A is disposed downstream of the fixing nip portion N in the conveying direction. The discharge unit **500** includes an air nozzle **501** configured to blow a compressed air for peeling the recording material S that has passed through the fixing nip portion N off from the fixing belt **301**, a lower separation guide **502** for peeling the recording material S off from the pressing roller **305**, and a pair of conveying rollers **503** configured to convey the peeled recording material S.

Belt Unit

[0031] FIG. **3** is a schematic view showing the belt unit. As shown in FIG. **3**, the belt unit **309** includes the endless fixing belt **301**, a heating roller **307**, a steering roller **308**, a fixing pad unit **300**, and belt unit side plates **313**. The belt unit side plates **313** are disposed to face each other on both end sides in the width direction (front-back direction) (a first support plate and a second support plate), and support one end portion and the other end portion of each of the heating roller **307**, the steering roller **308**, and a holding stay **302**.

[0032] The holding stay **302** of the fixing pad unit **300** is formed to be long enough to protrude outward from the left and right belt unit side plates **313** in the width direction, and the belt unit side plates **313** are fixed in the middle of the holding stay **302** of the fixing pad unit **300**. The frame side plates **311** are provided with fitting portions (not shown) into which both end portions of the holding stay **302** can be fitted. In addition, a fitting groove **700** is formed on the upper side of the belt unit side plate **313** on the downstream side (here, the left side) in the conveying direction. On the other hand, as shown in FIG. **2**, the frame side plate **311** is provided with a protruding portion **701** protruded to be fittable into the fitting groove **700**. In the case of the present embodiment, the belt unit **309** is moved from front to back with respect to the side plates **313**, that is, in the conveying direction (the arrow X direction), whereby the holding stay **302** is fitted into the fitting portions, and the protruding portions **701** are fitted into the fitting grooves **700**. In other words, the portions of the holding stay **302** that protrude outward from the belt unit side plates **313** in the width direction function as support portions that support the belt unit side plates **313** to the frame side plates **311**. The belt unit **309** is supported by the frame side plates **311** via the holding stay **302** and the protruding portions **701** as described above. Therefore, the heating roller **307** and the steering roller **308** are supported only by the belt unit side plates **313** and are not supported by the frame side plates **311**.

Fixing Belt

[0033] The fixing belt **301** is stretched by two stretching rollers of the heating roller **307** and the steering roller **308** and the fixing pad unit **300**. The fixing belt **301** has thermal conductivity, heat resistance, or the like, and is formed in a thin cylindrical shape. The fixing belt **301** has a three-layer structure in which a base layer, an elastic layer, and a release layer are formed on a base material such as rubber in order from the inner peripheral side. As an example, a polyimide resin (PI) having a thickness of "80 μm" is used as the base layer, silicone rubber having a thickness of "300 μm" is used as the elastic layer, and PFA having a thickness of "30 μm" is used as the release layer.

Heating Roller

[0034] The heating roller **307** serving as the stretching roller is disposed inside the fixing belt **301** and is rotatably pivoted by the belt unit side plates **313** so as to stretch the fixing belt **301** together with the fixing pad unit **300**. The heating roller **307** is a metal roller formed of, for example, a metal such as aluminum or stainless steel having a thickness of "1 mm" in a cylindrical shape, and a halogen heater **306** for heating the fixing belt **301** is disposed inside the heating roller **307**. The heating roller **307** is heated up to a predetermined temperature by the halogen heater **306**, whereby the fixing belt **301** is heated by the heating roller **307**. A gear (not shown) is fixed to the right end portion of the heating roller **307**, and the heating roller **307** is driven to rotate by a motor M**2** via a driving gear unit **322** that is installed in the fixing frame **8**A. The driving gear unit **322** serving as a driving transmission member transmits a driving force from the motor M**2** serving as a driving source to the heating roller **307** in an arrow W direction to rotate the heating roller **307**. The heating roller **307** and the pressing roller **305** may be driven to rotate by any one of the motor M**1** or the motor M**2** as a common driving source. For example, in a case where the motor M**1** is a common driving source, the driving gear unit **322** is rotated in the arrow W direction by the motor M**1**.

Fixing Pad Unit

[0035] The fixing pad unit **300** serving as a stretching member has the holding stay **302** and the pressurizing pad **303**, is disposed inside the fixing belt **301**, and is attached to the belt unit side plates **313**. The holding stay **302** is, for example, a stiff metal member that extends in the width direction of the fixing belt **301**, and holds the pressurizing pad **303**. The pressurizing pad **303** serving as a pad member is provided inside the fixing belt **301** in a non-rotating manner. The pressurizing pad **303** is a resin member that extends in the width direction, and the widthwise length of the pressurizing pad **303** is longer than the widthwise length of the recording material having the maximum size capable of forming an image. The pressurizing pad **303** is formed of, for example, a resin having good insulating properties and heat resistance, such as a liquid crystal polymer resin (LCP).

[0036] In order to reduce the frictional force between the fixing belt **301** and the pressurizing pad **303**, it is preferable that a sliding member (not shown) is provided between the pressurizing pad **303** and the fixing belt **301**. The sliding member is disposed at a position facing the pressing roller **305** with the fixing belt **301** interposed therebetween. In addition, it is preferable that a lubricant such as silicone oil is applied to the inner peripheral surface of the fixing belt **301** in order to smoothly slide the fixing belt **301** on the sliding member.

[0037] In the present embodiment, the fixing belt **301** is pressed against the pressurizing pad **303** held by the holding stay **302** from the outside by the pressing roller **305**. Thereby, a fixing nip portion N with a wide nip in which the length in the conveying direction and the widthwise length have been secured as described above is formed between the pressing roller **305** and the fixing belt **301**. In addition, the resin pressurizing pad **303** is held by the holding stay **302** made of a highly stiff metal, whereby deflection of the pressurizing pad **303** caused by the pressure during pressing is reduced, and a uniform fixing nip width in the width direction is obtained.

Steering Roller

[0038] In the belt unit **309**, due to the accuracy of a component, a so-called alignment error may occur in which the rotation axis directions of the stretching rollers that stretch the fixing belt **301** are not parallel and are out of sync. In that case, belt tracking in which the rotating fixing belt **301** moves aside the end portion in the width direction is likely to occur, and the fixing belt **301** or other members may be damaged. Therefore, in the present embodiment, in order to suppress belt tracking, one of the plurality of stretching rollers that stretch the fixing belt **301** is used as the steering roller **308**. The steering roller **308** is, for example, a member in which a rubber material for increasing the grip force with respect to the fixing belt **301** is provided on a surface of a hollow metal shaft (for example, a stainless steel shaft) formed in a diameter of "φ20 mm".

[0039] In the case of the present embodiment, the steering roller **308** is disposed downstream of the heating roller **307** and upstream of the pressurizing pad **303** in the rotation direction of the fixing belt **301**. The steering roller **308** controls the position of the rotating fixing belt **301** in the width direction. The steering roller **308** has a swing center at the center portion in the width direction and is provided such that the steering roller **308** can be tilted with respect to the fixing belt **301** by a steering mechanism, not shown. Since a tension difference is generated between both end portions of the fixing belt **301** depending on the tilting of the steering roller **308**, the fixing belt **301** moves in the width direction. As described above, in the present embodiment, in a case where the alignment error occurs due to the accuracy of a component as in the related art, belt tracking can be suppressed by the steering roller **308**.

[0040] By the way, in recent years, in order to handle an increase in the speed of image formation, the rotation speed of the fixing belt **301** may be increased, and the pressurizing force in the fixing nip portion N may be increased. In such a case, it is necessary to increase the driving torque of the heating roller **307**, and the rotational driving force that the heating roller **307** receives from the driving gear unit **322** is thus increased. However, depending on the magnitude of the rotational driving force, the driving gear unit **322** side of the belt unit **309** may receive a large force in an arrow F**2** direction (refer to FIG. **2**), and the belt unit side plates **313** on the driving gear unit **322** side (here, the right side) may deform. In a case where one of the left and right belt unit side plates **313** deforms, there is a concern that a larger alignment error may occur, and it is difficult to suppress belt tracking by the steering roller **308**.

[0041] In view of the above-described point, in the present embodiment, the operator is enabled to manually adjust the alignment between the stretching rollers in the belt unit **309** from the outside of the fixing frame **8**A to an extent that belt tracking can be suppressed by the steering roller **308**. Hereinafter, the pressing mechanism **401** for implementing the adjustment will be described using FIG. **2** and FIG. **4**. FIG. **4** is a cross-sectional view showing the pressing mechanism of the first embodiment.

Pressing Mechanism

[0042] As shown in FIG. **2** and FIG. **4**, the pressing mechanism **401** is composed of an abutting member **402**, a spring member **403**, an adjustment screw **404**, an adjustment frame **405**, attachment members **311***a*, and a rotation restricting member **311***b*. The pressing mechanism **401** is attached to the fixing frame **8**A via two attachment members **311***a* fixed to the fixing frame **8**A (specifically, the frame side plate **311**) in advance. The attachment members **311***a* are disposed side by side at an interval in the front-back direction. Through holes **311***c* are formed in the attachment members **311***a*, and the shaft portion of the adjustment screw **404** is inserted into these through holes **311***c*, whereby the adjustment screw **404** is supported by the attachment members **311***a* so as to be reciprocally movable in the front-back direction.

[0043] In the adjustment screw **404** supported by the two front and back attachment members **311***a*, the adjustment frame **405** is disposed between the attachment members **311***a*. The adjustment frame **405** is formed in, for example, a polygonal shape such as a hexagonal shape. A helical groove is formed in the shaft portion of the adjustment screw **404**, and a screw hole **405***a* that meshes with the shaft portion of the adjustment screw **404** is formed in the adjustment frame **405**. That is, the adjustment frame **405** is provided to allow the shaft portion of the adjustment screw **404** to move by the rotation of the adjustment screw **404**.

[0044] In the present embodiment, the rotation restricting member **311***b* is fixed to the fixing frame **8**A in the same manner as the attachment members **311***a*. The rotation restricting member **311***b* abuts on one surface of the outer peripheral surface of the adjustment frame **405**, and restricts the rotation of the adjustment frame **405** such that the adjustment frame **405** does not rotate together with the adjustment screw **404** in a case where the adjustment screw **404** is rotated. In this way, the adjustment frame **405** allows the shaft portion of the adjustment screw **404** to reciprocally move within a range H in the front-back direction of the rotation restricting member **311***b* by the rotation of the adjustment screw **404**. Accordingly, the position of the adjustment frame **405** changes on the shaft portion of the adjustment screw **404** by the rotation of the adjustment screw **404**. For example, in a case where the adjustment screw **404** is rotated clockwise, the adjustment frame **405** moves from the back side to the front side with respect to the shaft portion, and in a case where the adjustment screw **404** is rotated counterclockwise, the adjustment frame **405** moves from the front side to the back side with respect to the shaft portion.

[0045] In addition, in a state where the shaft portion of the adjustment screw **404** has been inserted into the front and back attachment members **311***a*, the spring member **403** (a first elastic member and a second elastic member) inserted into the shaft portion of the adjustment screw **404** are disposed between the adjustment frame **405** and the front attachment member **311***a*. The spring member **403** is provided to urge the abutting member **402** toward the belt unit side plate **313**, and the adjustment screw **404** can adjust the urging force of the spring member **403**.

[0046] In the case of the present embodiment, one end of the spring member **403** is connected to the adjustment frame **405**, the other end of the spring member **403** is connected to the front attachment member **311***a*, and the spring member **403** applies a force of pulling the adjustment frame **405** to the front side with reference to the front attachment member **311***a*. As described above, in a case where the adjustment frame **405** moves with respect to the shaft portion of the adjustment screw **404** by the rotation of the adjustment screw **404**, the interval between the adjustment frame **405** and the front attachment member **311***a* changes, and the spring member **403** thus expands and contracts. The abutting member **402** (a first abutting unit and a second abutting unit) that abuts on the belt unit side plate **313** is provided at the back-side tip end of the adjustment screw **404**. Therefore, the force of pulling the adjustment frame **405** to the front side changes according to the expansion and contraction of the spring member **403**, whereby the force of pushing the abutting member **402** from the front side toward the back side changes. For example, in a case where the interval between the adjustment frame **405** and the front attachment member **311***a* has been narrowed, and the spring member **403** has contracted, a pressing force of the abutting member **402** that press the belt unit side plate **313** becomes large. On the other hand, in a case where the interval between the adjustment frame **405** and the front attachment member **311***a* has been widened, and the spring member **403** has expanded, the pressing force of the abutting member **402** that press the belt unit side plate **313** becomes small. The belt unit side plate **313** deform by being pressed by the abutting member **402**.

[0047] The pressing mechanism **401** is disposed at each of the left and right positions at which the pressing mechanisms **401** can press each of the left and right belt unit side plates **313** from the front side (a first pressing mechanism and a second pressing mechanism). In the present embodiment, one of the two pressing mechanisms **401** is capable of pressing the left belt unit side plate **313** in a first direction intersecting with the rotation axis direction of the heating roller **307**, and the other pressing mechanism **401** is capable of pressing the right belt unit side plate **313** in a second direction intersecting with the rotation axis direction of the heating roller **307**. The first direction and the second direction may be the same direction.

[0048] As shown in FIG. **2**, in the cover member **312**, adjustment openings **406** (a first opening portion and a second opening portion) communicating with the outside are formed at left and right positions overlapping with the head portions of the adjustment screws **404** (a first adjustment unit and a second adjustment unit) as viewed from the front side. In the present embodiment, the adjustment opening **406** may further include a lid that is fitted with the adjustment opening **406** in the cover member **312** and suppresses contact with the outside. The lid is formed of the same material as the cover member **312** and can be opened and closed for an adjustment operation. The adjustment opening **406** is formed to be large enough for a tool for rotating the adjustment screw **404** to be inserted in order to access the adjustment screw **404** in the fixing frame **8**A. The operator can insert, for example, a driver from the adjustment opening **406** and rotate the adjustment screw **404** with the driver. The operator can adjust the pressing force of the pressing mechanism **401** against the belt unit side plate **313** by changing the rotation direction of the adjustment screw **404**. The pressing force of the pressing mechanism **401** installed at each of the left and the right is adjusted, whereby the amount of deformation of each of the left and right belt unit side plates **313** changes.

[0049] In the cover member **312**, no openings are formed in a portion that connects the adjustment openings **406** (the first opening portion and the second opening portion) formed on the left and the right, respectively. That is, in the cover member **312**, a wall portion **3121** that separates the inside and the outside of the cover member **312** is provided between the left and right adjustment openings **406** (refer to FIG. **10**). The adjustment openings **406** are provided to enable the operator to operate the pressing mechanism **401** even in a state where the cover member **312** is closed. However, the thermal insulating performance, which is the intrinsic purpose of the cover member **312**, deteriorates as much as the openings being provided. Therefore, the opening formed in the cover member **312** should be minimized, and the opening is not formed in the portion that connects the left and right adjustment openings **406**. In other words, in the cover member **312**, a portion between the left and right adjustment openings **406** (a portion between the first opening portion and the second opening portion) is blocked.

[0050] As described above, in the present embodiment, the pressing mechanism **401** that can be operated by the operator from the outside of the fixing frame **8**A is provided. In addition, the pressing force of the pressing mechanism **401** with respect to the belt unit side plate **313** is adjusted in accordance with the operator's operation, whereby each of the left and right belt unit side plates **313** can be deformed. Each of the left and right belt unit side plates **313** is deformed as described above, whereby the alignment between the heating roller **307** and the steering roller **308** supported by the belt unit side plates **313** is adjusted. The alignment adjustment by deforming the belt unit side plate **313** needs to be performed to an extent that belt tracking can be suppressed by the steering roller **308** after the adjustment. The operator can operate the pressing mechanism **401** from the outside of the fixing frame **8**A, and thus can easily adjust the alignment.

Second Embodiment

[0051] Next, a second embodiment will be described using FIG. **5** and FIG. **6**. FIG. **5** shows a fixing device of the second embodiment, and FIG. **6** shows a pressing mechanism of the second embodiment. A fixing device **80** of the second embodiment shown in FIG. **5** is different from the fixing device **8** of the first embodiment shown in FIG. **2** only in the configuration of a pressing mechanism **410**, and the other configurations are the same. Therefore, the same configuration as that of the fixing device **8** according to the first embodiment described above will be denoted by the same reference numeral, and the description thereof will be simplified or omitted.

[0052] As shown in FIG. **5** and FIG. **6**, the pressing mechanism **410** is composed of an adjustment screw **411**, a swing lever **412**, a spring member **413**, and an attachment member **321**. In the second embodiment, as in the first embodiment, the pressing mechanism **410** is disposed at each of the left and right positions at which the pressing mechanisms **410** can press each of the left and right belt unit side plates **313** from the front side. The pressing mechanism **410** is attached to the fixing frame **8**A via the attachment member **321** fixed to the fixing frame **8**A (specifically, the frame side plates **311**) in advance and the swing lever **412**. The swing lever **412** is supported to be capable of swinging around a swing shaft **412***a* provided to protrude from the frame side plate **311**.

[0053] A helical groove is formed in the shaft portion of the adjustment screw **411**, and a screw hole **321***a* that meshes with the shaft portion of the adjustment screw **411** is formed in the attachment member **321**. That is, the adjustment screw **411** is reciprocally movable in the front-back direction with respect to the attachment member **321** by rotation. A through hole **412***c* is formed in the swing lever **412**, and the shaft portion of the adjustment screw **411** is inserted into this through hole **412***c*. In addition, in a state where the shaft portion of the adjustment screw **411** has been inserted into the through hole **412***c* in the swing lever **412**, the spring member **413** inserted into the shaft portion of the adjustment screw **411** is disposed between the swing lever **412** and the head portion of the adjustment screw **411**. The spring member **413** is connected to the swing lever **412** at one end and connected to the head portion of the adjustment screw **411** at the other end and is provided to swing the swing lever **412** back and forth.

[0054] The swing lever **412** is provided with a protruding portion **414** that protrudes backward between the through hole **412***c* and the swing shaft **412***a*. This protruding portion **414** abuts on the belt unit side plate **313**. In a case where the adjustment screw **421** has been rotated and moved backward, since the protruding portion **414** is in a state of abutting on the belt unit side plate **313**, the interval between the swing lever **412** and the head portion of the adjustment screw **411** is narrowed, and the spring member **413** contracts. In a case where the spring member **413** contracts, a force of pushing the swing lever **412** backward becomes large, and the pressing force of the protruding portion **414** that presses the belt unit side plate **313** thus becomes larger. On the other hand, in a case where the adjustment screw **414** is rotated and moved forward, the interval between the swing lever **412** and the head portion of the adjustment screw **411** is widened, and the spring member **413** expands. In a case where the spring member **413** expands, the force of pushing the swing lever **412** backward becomes small, and a pressing force of the protruding portion **414** that presses the belt unit side plate **313** thus becomes larger. In this way, the belt unit side plate **313** deform by being pressed by the protruding portion **414** of the swing lever **412**.

[0055] As shown in FIG. **5**, in the cover member **312**, the adjustment opening **406** is formed at a position overlapping with the head portion of the adjustment screw **411** as viewed from the front side. Therefore, the operator can insert, for example, a driver from the adjustment opening **406** and rotate the adjustment screw **411** with the driver. The operator can change the pressing force of the pressing mechanism **410** against the belt unit side plate **313** by changing the rotation direction of the adjustment screw **411**. The pressing forces of the pressing mechanisms **410** installed at the left and the right, respectively, are changed, whereby the amounts of deformation of the left and right belt unit side plates **313** changes, respectively. Accordingly, the alignment between the heating roller **307** and the steering roller **308** supported by the belt unit side plates **313** is adjusted. The operator can operate the pressing mechanism **410** from the outside of the fixing frame **8**A, and thus can easily adjust the alignment.

Third Embodiment

[0056] Next, a third embodiment will be described using FIG. **7** and FIG. **8**. FIG. **7** shows a fixing device of the third embodiment, and FIG. **8** shows a pressing mechanism of the third embodiment. A fixing device **800** of the third embodiment shown in FIG. **7** is different from the fixing device **8** of the first embodiment shown in FIG. **2** only in the configuration of a pressing mechanism **420**, and the other configurations are the same. Therefore, the same configuration as that of the fixing device **8** according to the first embodiment described above will be denoted by the same reference numeral, and the description thereof will be simplified or omitted.

[0057] As shown in FIG. **7** and FIG. **8**, the pressing mechanism **420** is composed of the adjustment screw **421**, an adjustment frame **422**, a spring member **423**, a swing lever **424**, an attachment member **331**, and a rotation restricting member **332**. In the third embodiment, as in the first embodiment, the pressing mechanism **420** is disposed at each of the left and right positions at which the pressing mechanisms **420** can press each of the left and right belt unit side plates **313** from the front side. The pressing mechanism **420** is attached to the fixing frame **8**A via two attachment members **331** fixed to the fixing frame **8**A (specifically, the frame side plates **311**) in advance. The attachment members **331** are disposed side by side at an interval in the front-back direction. Through holes **331***a* are formed in the attachment members **331**, and the shaft portion of the adjustment screw **421** is inserted into these through holes **331***a,* whereby the adjustment screw **421** is supported by the attachment members **331** so as to be reciprocally movable in the front-back direction.

[0058] In the adjustment screw **421** supported by the two front and back attachment members **331**, the adjustment frame **422** is disposed between the attachment members **331**. The adjustment frame **422** is formed in, for example, a polygonal shape such as a hexagonal shape. A helical groove is formed in the shaft portion of the adjustment screw **421**, and a screw hole **422***a* that meshes with the shaft portion of the adjustment screw **421** is formed in the adjustment frame **422**. That is, the adjustment frame **422** is provided to allow the shaft portion of the adjustment screw **421** to move by the rotation of the adjustment screw **421**.

[0059] In the present embodiment, the rotation restricting member **332** is fixed to the fixing frame **8**A in the same manner as the attachment members **331**. The rotation restricting member **332** abuts on one surface of the outer peripheral surface of the adjustment frame **422**, and restricts the rotation of the adjustment frame **422** such that the adjustment frame **422** does not rotate together with the adjustment screw **421** in a case where the adjustment screw **421** is rotated. Accordingly, along with the rotation of the adjustment screw **421**, the adjustment frame **422** can reciprocally move the shaft portion of the adjustment screw **421**, and the position of the adjustment frame **422** on the shaft portion of the adjustment screw **421** changes. For example, in a case where the adjustment screw **421** is rotated clockwise, the adjustment frame **422** moves from the back side to the front side with respect to the shaft portion, and in a case where the adjustment screw **421** is rotated counterclockwise, the adjustment frame **422** moves from the front side to the back side with respect to the shaft portion.

[0060] One end of the spring member **423** is connected to, among the outer peripheral surfaces of the adjustment frame **422**, a surface opposite to the surface regulated by the rotation restricting member **332**. In the present embodiment, the swing lever **424** is supported to be capable of swinging around a swing shaft **424***a* provided to protrude from the frame side plate **311**. The other end of the spring member **423** is connected to one end side of the swing lever **424**. That is, the spring member **423** is stretched by the adjustment frame **422** and the swing lever **424**. In addition, the swing lever **424** is provided with an abutting member **425** on the other end side opposite to one end side to which the spring member **423** is connected. The abutting member **425** abuts on the belt unit side plate **313**.

[0061] In a case where the adjustment frame **422** has moved forward with respect to the shaft portion of the adjustment screw **421** by the rotation of the adjustment screw **421**, since the abutting member **425** is in a state of abutting on the belt unit side plate **313**, the spring member **423** expands. In a case where the spring member **423** expands, since the force of pulling one end side of the swing lever **424** to the front side becomes large, the pressing force of the abutting member **425** on the other end side that presses the belt unit side plates **313** becomes large. On the other hand, in a case where the adjustment frame **422** has moved backward by the rotation of the adjustment screw **421**, the spring member **423** contracts. In a case where the spring member **423** contracts, since the force of pulling one end side of the swing lever **424** to the front side becomes small, the pressing force of the abutting member **425** on the other end side that presses the belt unit side plate **313** becomes small. In this way, the belt unit side plate **313** deform by being pressed by the abutting member **425** of the swing lever **424**.

[0062] As shown in FIG. **7**, in the cover member **312**, the adjustment opening **406** is formed at a position overlapping with the head portion of the adjustment screw **421** as viewed from the front side. Therefore, the operator can insert, for example, a driver from the adjustment opening **406** and rotate the adjustment screw **421** with the driver. The operator can change the pressing force of the pressing mechanism **420** against the belt unit side plate **313** by changing the rotation direction of the adjustment screw **421**. The pressing forces of the pressing mechanisms **420** installed at the left and the right, respectively, are changed, whereby the amounts of deformation of the left and right belt unit side plates **313** changes, respectively. Accordingly, the alignment between the heating roller **307** and the steering roller **308** supported by the belt unit side plates **313** is adjusted. The operator can operate the pressing mechanism **420** from the outside of the fixing frame **8**A, and thus can easily adjust the alignment.

Other Embodiments

[0063] In each of the above-described embodiments, the operator inserts a tool such as a driver from the adjustment opening **406** and rotates the adjustment screw (**404**, **411**, or **421**), but the present invention is not limited thereto. For example, the operator may be enabled to grip the adjustment screw (**404**, **411**, or **421**) by hands and rotate the adjustment screw (**404**, **411**, or **421**). In order for that, at least a part of the adjustment screw (**404**, **411**, or **421**) may be exposed to the outside of the cover member **312** through the adjustment opening **406**. In addition, as shown in FIG. **9**, for example, it is preferable that a grip portion **900** is provided at a portion of the adjustment screw **404** that is exposed to the outside of the cover member **312**, and the operator can operate the adjustment screw (**404**, **411**, or **421**) to rotate by gripping the grip portion **900**.

[0064] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0065] This application claims the benefit of Japanese Patent Application No.2024-035941, filed Mar. 8, 2024, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. A fixing device that fixes a toner image formed on a recording material to the recording material, comprising:

a belt unit including:
a rotating endless belt;
a stretching roller disposed inside the belt and configured to stretch the belt;
a first support plate supporting one end portion of the stretching roller in a rotation axis direction; and
a second support plate supporting the other end portion of the stretching roller in the rotation axis direction;
a rotatable pressing member disposed outside the belt and forming a nip portion capable of sandwiching and conveying the recording material together with the belt;
a frame supporting the first support plate and the second support plate;
an adjustment unit provided on the frame and being capable of adjusting a position of the first support plate in a direction intersecting with the rotation axis direction; and
a cover member covering the belt unit, the frame, and the adjustment unit,
wherein the cover member includes an opening portion formed at a position corresponding to the adjustment unit and communicating with an outside.

2. The fixing device according to claim **1**, wherein the adjustment unit is configured to adjust the position of the first support plate by pressing the first support plate in the direction intersecting with the rotation axis direction.

3. The fixing device according to claim **2**, wherein the adjustment unit further includes
an operation unit receiving an operation of adjusting a pressing force of the adjustment unit,
an abutting unit abutting on the first support plate, and
an elastic member for urging the abutting unit toward the first support plate, and the operation unit is capable of adjusting an urging force of the elastic member.

4. The fixing device according to claim **3**, wherein the operation unit can be operated from an outside of the cover member by a tool inserted through the opening portion.

5. The fixing device according to claim **3**, wherein the operation unit is at least partially exposed to an outside of the cover member through the opening portion and can be operated on the outside of the cover member.

6. The fixing device according to claim **1**, wherein the belt unit further includes a steering roller configured to stretch the belt together with the stretching roller inside the belt and control a position of the belt in the rotation axis direction by tilting with respect to the rotation axis direction.

7. The fixing device according to claim **6**, wherein the steering roller is supported by the first support plate and the second support plate such that the steering roller can be tilted with respect to the rotation axis direction.

8. The fixing device according to claim **1**, further comprising:
a stretching member disposed inside the belt and configured to stretch the belt together with the stretching roller,
wherein the stretching member includes
a pad member abutting on an inner peripheral surface of the belt, and
a stay supporting the pad member,
both end portions of the stay in the rotation axis direction are fixed to the first support plate and the second support plate, and
portions of the stay that protrude outward from the first support plate and the second support plate in the rotation axis direction are support portions supporting the first support plate and the second support plate on the frame.

9. The fixing device according to claim **1**, wherein the cover member includes an openable lid configured to fit into the opening portion.

10. The fixing device according to claim **1**, further comprising:
a second adjustment unit provided on the frame and being capable of adjusting a position of the second support plate in the direction intersecting with the rotation axis direction,
wherein the cover member further includes a second opening portion formed at a position corresponding to the second adjustment unit and communicating with the outside.

11. The fixing device according to claim **10**, wherein the cover member includes a wall portion separating an inside and an outside of the cover member between the opening portion and the second opening portion.

* * * * *